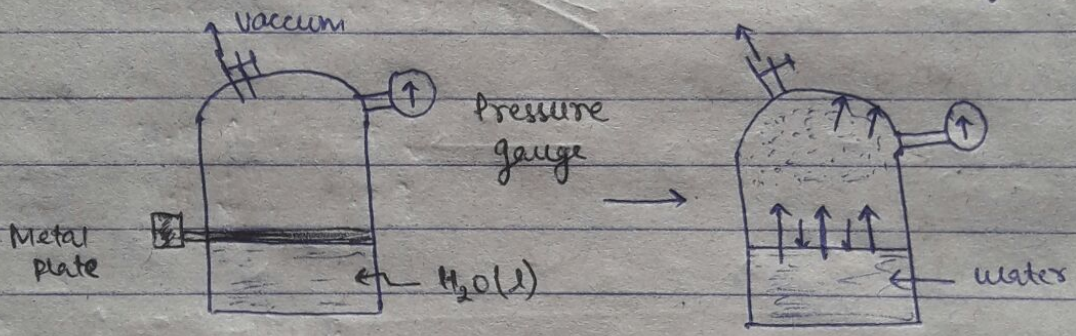
LIQUID SOLUTION

Liquid Solution and Colligative properties :

Liquid Solution
- Liquid in liquid (Raoult's law)
- Gas in liquid (Henry's law)
- Solids in liquid (colligative properties of solution)

Vapour pressure :

It is the pressure exerted by the vapours of a substance (liquid or solid) at equilibrium state at a given temperature.

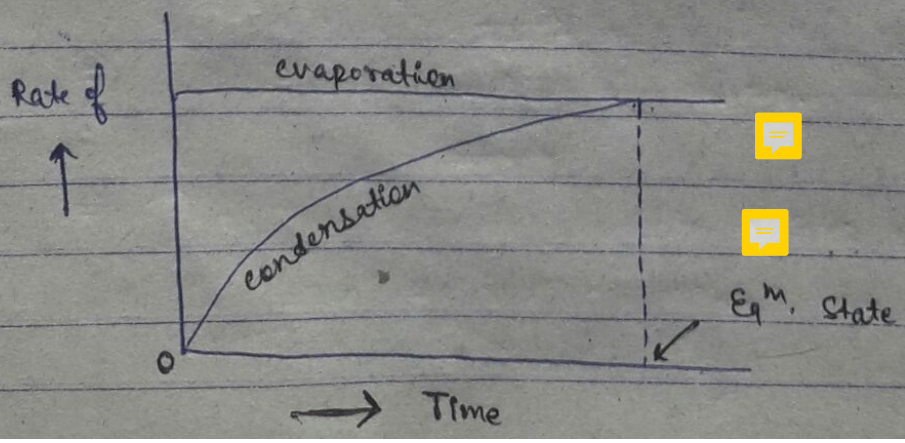


Metal plate withdrawn.

$$H_2O(l) \underset{\text{condensation}}{\overset{\text{evaporation}}{\rightleftharpoons}} H_2O(g)$$

At eqm. state

Rate of evaporation = Rate of condensation



vapour pressure defined for solids too

* Features of Vapour Pressure :-

(1) V.P of a given substance does not depend on surface area and amount of substance.

(2) V.P of a substance depends on temperature.

$T \uparrow$, V.P $\uparrow$

→ Variation of V.P with temperature is expressed by Clausius - Clapeyron equation.

$$\ln \frac{P_2}{P_1} = \frac{\Delta H_{vap}}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

P_1, P_2 → V.P of a liquid at temp. T_1, T_2

ΔH_{vap} → Molar enthalpy of vapourisation.

$$\Rightarrow \log \frac{P_2}{P_1} = \frac{\Delta H_{vap}}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\ln P = \ln k - \frac{\Delta H_{vap}}{RT} \quad [\text{where } k \rightarrow \text{constant}]$$

$$\frac{d(\ln P)}{dT} = 0 - \frac{\Delta H_{vap}}{R} \times \left(\frac{-1}{T^2} \right)$$

$$\Rightarrow d(\ln P) = \frac{\Delta H_{vap}}{RT^2} dT$$

→ differential form of clausius - clapeyron equation.

⇒ with increase in temperature, V.P of that liquid is more sharply which has greater $\Delta H_{vap.}$

OR

V.P of a liquid is more sensitive towards temp. variation for that liquid which has high $\Delta H_{vap.}$

(3) Stronger are the intermolecular forces, lower will be the escaping tendency of molecules and hence lower will be the V.P

eg: water is less volatile than ethanol, due to stronger H-bonding in water.

If V.P ↑, B.P ↓
If V.P ↓, B.P ↑

* Boiling Point:
The temperature at which vapour pressure of a liquid becomes equal to the external pressure.

* Normal boiling point (NBP):
The temperature at which vapour pressure of a liquid becomes equal to 1 atm.

⇒ 100°C is the NBP of water.

Imp ⇒ There are infinite possible values of B.P of a liquid depending upon external pressure.

(4) Relative humidity (R.H)

$$= \frac{P_{H_2O}(vap)}{\text{V.P of } H_2O(l)} \times 100$$

Both pressure are taken at same temp. condition

↑ Constant at a given temperature

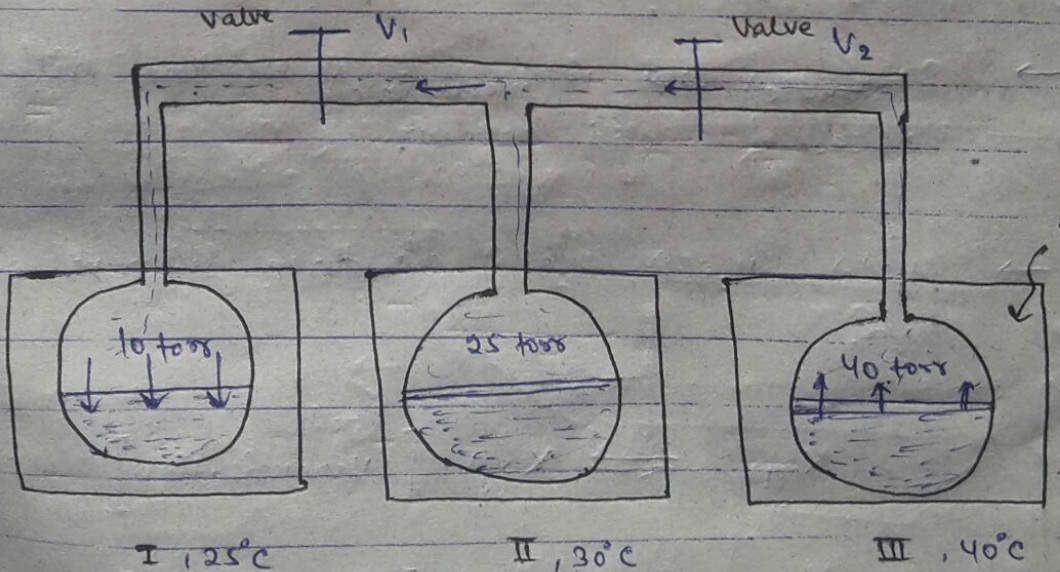
* V.P of water is ~~called~~ also called as aqueous tension.

When R.H = 100%,

$P_{H_2O}(vap)$ = V.P of H_2O

(Dynamic eqm exists b/w water vapours and water liquids)

→ Such air or gas with RH = 100% is called Saturated air or saturated gas.



Q. ***** 5 Mrs question

→ I, II and III contain water

→ $(VP)_{H_2O(l)}$ at 25°C = 10 torr
30°C = 25 torr
40°C = 40 torr

} given data.

Valves V1 and V2 are opened. Find the final partial pressures and amount of water in all the vessels

Ans. Finally at eq^m,

Vessel - II and III will not have any liquid water and will contain only water vapours at 10 torr.

⇒ vessel - I will have larger amount of liquid water and pressure of water vapours above vessel - I will be 10 torr.

* Clayperon equation (For phase change)

Phase : Part of a substance which has uniform composition and uniform properties throughout.

→ diamond and graphite are two different phases of solid carbon.

→ $S_{(rhombic)}$ and $S_{(monoclinic)}$ are two different phases of solid sulphur.

→ $H_2O(s)$, $H_2O(l)$, $H_2O(g)$ are three different phases of H_2O.

$$X_{(L)} \rightleftharpoons X_{(g)} \quad [\text{Vapourisation}]$$

$$X_{(s)} \rightleftharpoons X_{(g)} \quad [\text{Sublimation}]$$

$A \rightleftharpoons B$, A & B are two different phases.

At temp. (T & P) G_A G_B

At eq^m. $G_A = G_B \Rightarrow (\Delta G_T)_{T,P} = 0$

At temp : $A \rightleftharpoons B$

$(T+dT), (P+dP)$	$G_A + dG_A$	$G_B + dG_B$

At eqm. $(\Delta G)_{T,P} = 0$

$G_A + dG_A = G_B + dG_B$

$dG = VdP - SdT$ for any substance

$V_A \times dP - S_A \times dT = V_B dP - S_B \times dT$

$dP(V_A - V_B) = (S_A - S_B)dT$

$$\frac{dP}{dT} = \frac{S_A - S_B}{V_A - V_B} = \frac{S_B - S_A}{V_B - V_A}$$

$$\boxed{\frac{dP}{dT} = \frac{\Delta S}{\Delta V}}$$ → ~~Clayp~~ clapeyron equation.

Applying this eq (1) for vaporisation,

$H_2O(l) \rightleftharpoons H_2O(g)$

$$\frac{dP}{dT} = \frac{\Delta S}{V_g - V_l}, \quad V_l << V_g$$

$$\Delta S = \frac{\Delta H_{vap}}{T}$$

$$= \frac{\Delta H_{vap}}{T \times \frac{RT}{P}}, \quad V_g = \frac{RT}{P} \text{ for 1 mol}$$

$$\Rightarrow \int_{P_1}^{P_2} \frac{dP}{P} = \int_{T_1}^{T_2} \frac{\Delta H_{vap}}{RT^2} dT$$

taken constant.

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_{vap}}{R} \left\{ -\frac{1}{T} \right\}_{T_1}^{T_2}$$

$$\ln \frac{P_2}{P_1} = \frac{\Delta H_{vap}}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$ → clausius - clapeyron equation

where P_1, P_2 are V.P of a liquid at temp T_1 and T_2

ΔH_{vap} → Molar enthalpy of vapourisation of liquid

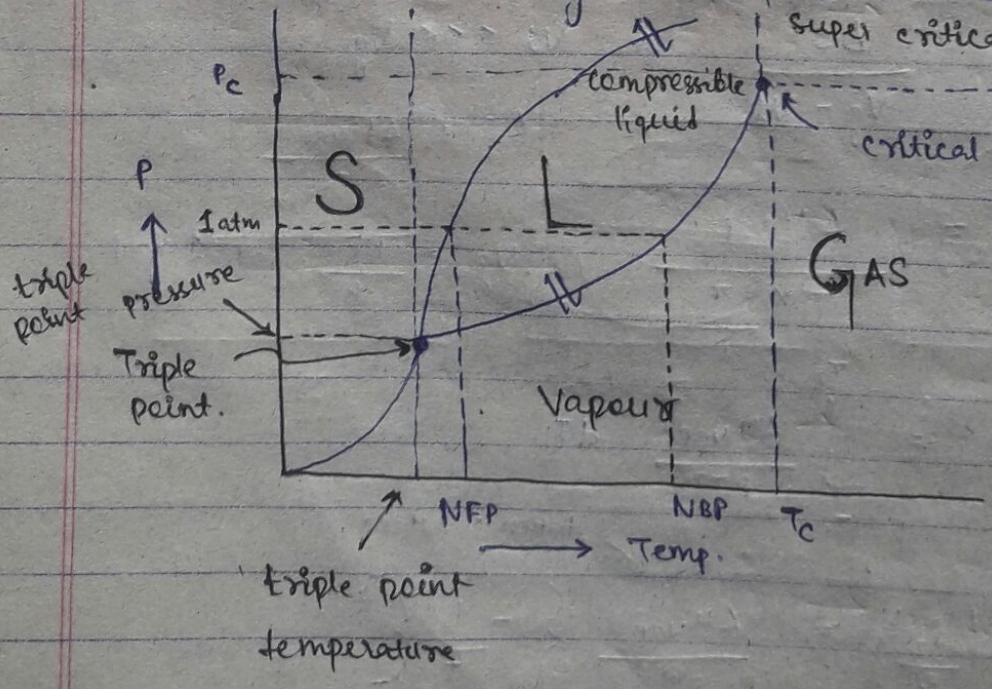
T↑, V.P↑

* Phase diagram (of a Pure substance)

→ It is P Vs T diagram of a substance which helps us to predict or find the stable phase of substance at given P & T condition.

→ Phase boundary :

A solid line in phase diagram which separates two phases ~~And~~ At phase boundary, two phases are in equilibrium.



NBP = Normal Boiling Point
NFP = Normal Freezing Point

(General Phase Diagram)

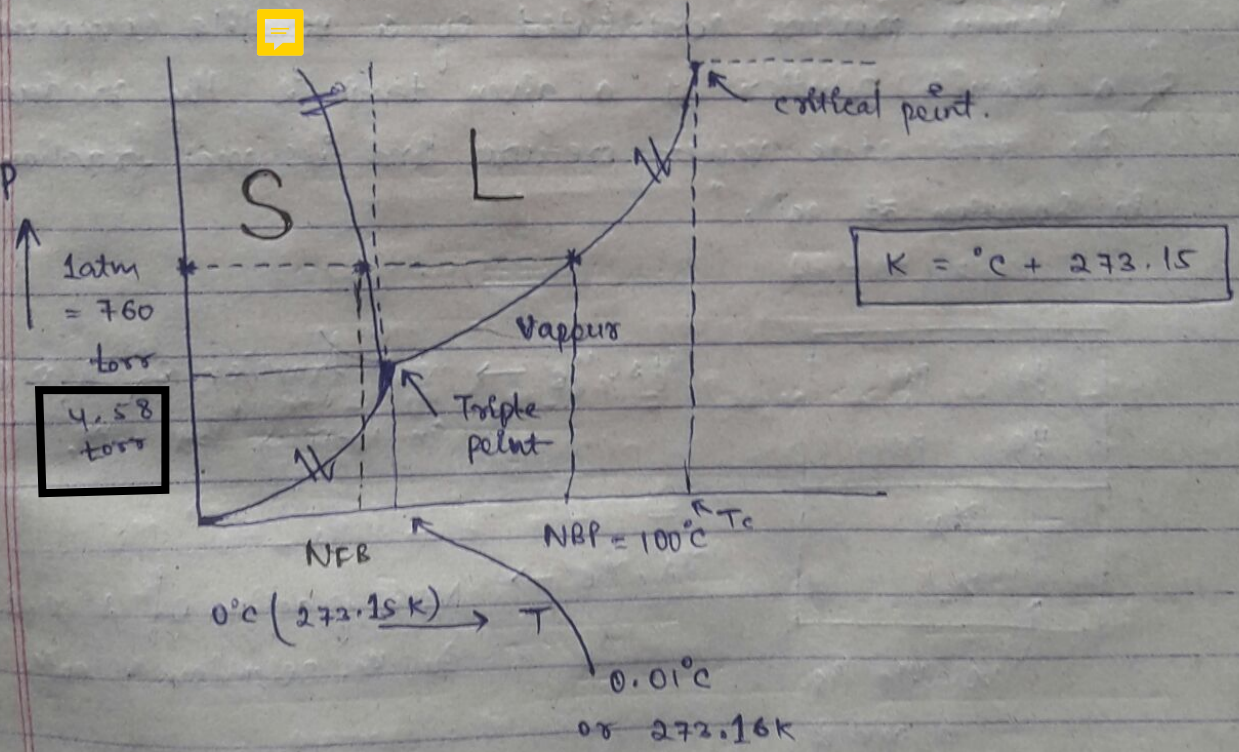
Triple Point :

Characteristic points in phase diagram of a substance where three phases are in equilibrium with each other.

→ There may be more than one triple point in phase diagram of a substance.

→ At triple point, S, L, G in eq^{m}. or two 's' and one liquid or one solid & two liquids may be in equilibrium.

* Phase diagram of H_2O :



$K = °C + 273.15$

Clapeyron eq^{n},

$$\frac{dP}{dT} = \frac{\Delta S}{\Delta V}$$

→ For solid-liquid boundary,

$\Delta V = V_l - V_s$ ⇒ For ice, $d_{H_2O(l)} > d_{H_2O(s)}$

⇒ $V_l < V_s$

⇒ For ice, $d_{H_2O(l)} > d_{H_2O(s)}$

$\therefore (\Delta V)_{H_2O(s) \longrightarrow H_2O(l)} = -ve$

$\frac{dP}{dT} = -ve$

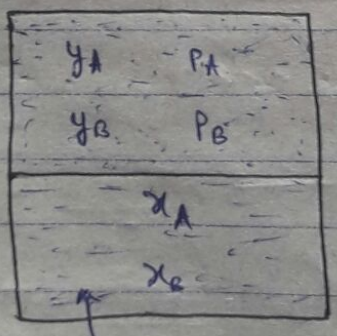
↑ slope of P vs T curve

* Solution of liquid in liquid (Raoult's law) (for ideal solution)

The solution which obeys Raoult's law in all range of concentration is called ideal solution.

According to Raoult's law,

" Vapour pressure contributed by a volatile component of a liquid solution at a given temp. is the product its pure state vapour pressure and its mol fraction in solution at eqm. "



y_A, y_B → mol fraction of A & B in vapour at eqm.

x_A, x_B → Mole fraction of A & B in liq. solution at eqm.

P_A, P_B → V.P contributed by A and B

$\mathring{P}A, \mathring{P}B$ → V.P of pure A and pure B.

As per Raoult's law,

$$P_A = x_A \times P_A^\circ$$
$$P_B = x_B \times P_B^\circ$$

Total V.P, $P_T = P_A + P_B$

$$\boxed{P_T = x_A P_A^\circ + x_B P_B^\circ}$$

Using dalton's law of partial pressure,

$$P_A = y_A \times P_T$$
$$P_B = y_B \times P_T$$

Combining Raoult's law & Dalton's law,

$$\boxed{x_A P_A^\circ = y_A \times P_T}$$

$$\boxed{x_B P_B^\circ = y_B \times P_T}$$

we are here dealing with a two phase system (liquid phase and vapour phase)

→ In a given phase, sum of mole fraction of all components = 1.

$$\Rightarrow x_A + x_B = 1 \quad | \quad y_A + y_B = 1$$

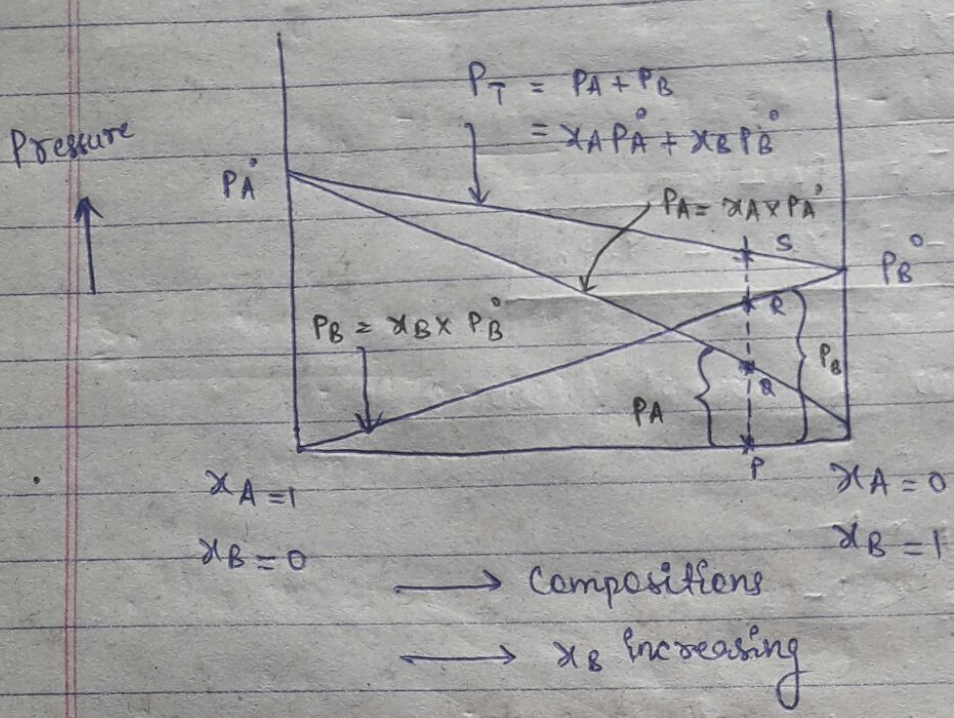
$$\frac{y_A \times P_T}{P_A^\circ} + \frac{y_B \times P_T}{P_B^\circ} = 1$$

$$\boxed{\frac{1}{P_T} = \frac{y_A}{P_A^\circ} + \frac{y_B}{P_B^\circ}}$$

Graphical Representation :-

All are straight lines here in Raoult's law.

Temp → constant in the graph.



→ Compositions

→ x_B increasing

$PQ = P_A$
$PR = P_B$
$PS = P_T$
$RS = PQ = P_A$

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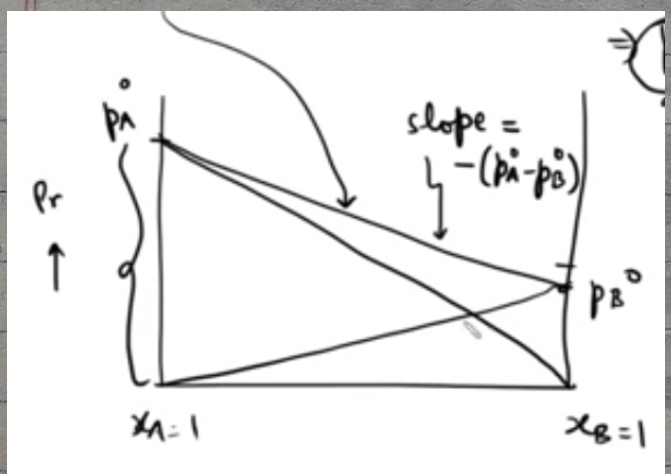
$PR = p_B$
$PS = p_T$
$RS = PQ = p_A$

$$P_T = x_A P_A^\circ + x_B P_B^\circ = (1 - x_B) P_A^\circ + x_B P_B^\circ$$

$$\Rightarrow \underbrace{P_T}_{y} = P_A^\circ - (P_A^\circ - P_B^\circ)\underbrace{x_B}_{x}$$



$y = mx + c$

Intercept = P_A°

Slope = $-(P_A^\circ - P_B^\circ)$

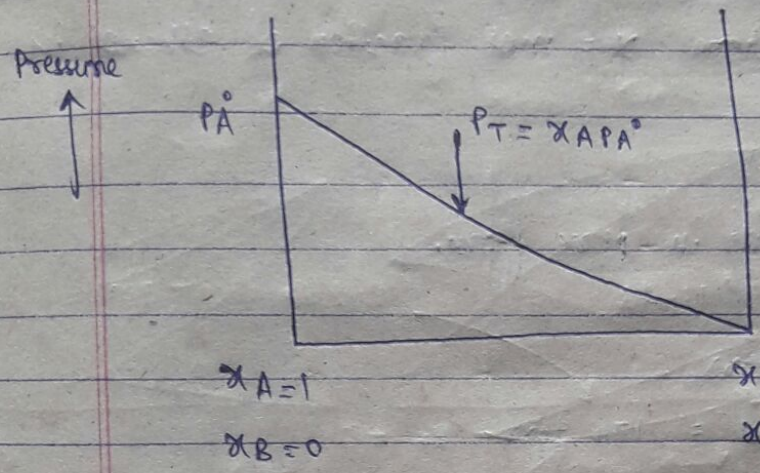
= -ve

If one of the component of solution is non-volatile

$$P_T = \chi_A P_A^\circ + \chi_B P_B^\circ$$

let B → non-volatile, $P_B^\circ = 0$

$$\Rightarrow \quad P_T = \chi_A P_A^\circ$$



Q. Two volatile liquids A and B are taken in 1:2 mass ratio to form solution. Find

(i) V.P of solution

(ii) Composition of vapour at eqm.

Given $P_A^\circ = 50$ torr, $P_B^\circ = 100$ torr

A → 40, B → 100 amu (Molar Mass)

Soln

A + B

$\frac{1}{40} : \frac{2}{100}$ Molar ratio $\Rightarrow \frac{1}{4} : \frac{1}{5}$

5 : 4 (Molar Ratio)

$\frac{5}{9}, \frac{4}{9}$

(i) $P_T = \chi_A P_A^\circ + \chi_B P_B^\circ$

$= \frac{5}{9} \times 50 + \frac{4}{9} \times 100$

$= \frac{250 + 400}{9} = \frac{650}{9}$ torr.

(b) Vapour composition :

$$y_A \times \frac{650}{9} = \frac{5}{9} \times 50$$

$$y_A = \frac{5}{13}, \quad y_B = 1 - \frac{5}{13} = \frac{8}{13}$$

Q. V.P of a solution of methanol - ethanol is expressed as $P = (170 - 60x)_{torr}$ where x = mol fraction of ethanol in solution.

Find V.P of both components in pure form.

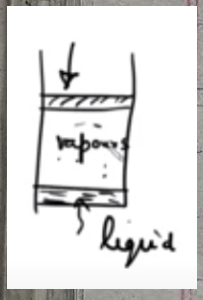
Soln : $P = 170 - 60x$ torr

If $x = 0$, P° methanol = 170 torr

If $x = 1$, P° ethanol = $170 - 60 \times 1$

= 110 torr

Q. One mol each of liquids A & B are mixed in a piston type vessel and piston is moved slowly and isothermally ($P_A^\circ = 75$ torr, $P_B^\circ = 25$ torr)



(a) Find pressure when first bubble is formed and also find composition of first bubble.

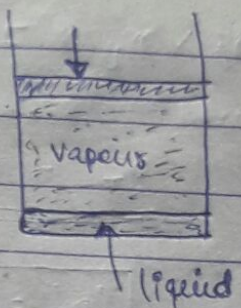
(b) Find pressure when last drop of liquid is remaining.

(c) Find pressure range in which both liquid & vapour forms co-exist.

(d) If out of 2 mol mixture, one mol solution is vapourised at pressure 'p', Find P.

Soln :-

(a) First bubble formed



$x_A = \frac{1}{2}$, $x_B = \frac{1}{2}$

$P_T = \frac{1}{2} \times 75 + \frac{1}{2} \times 25 = 50$ torr

$y_A \times 50 = \frac{1}{2} \times 75 \Rightarrow y_A = \frac{3}{4}$, $y_B = \frac{1}{4}$

composition of first bubble.

(b) Last drop of liquid

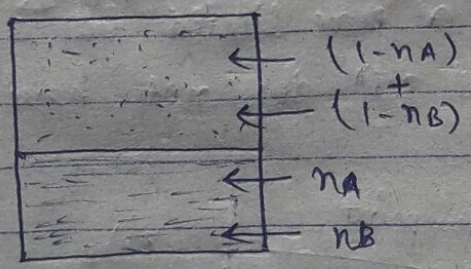
$y_A = \frac{1}{2}$, $y_B = \frac{1}{2}$, $\frac{1}{P_T} = \frac{1}{2 \times 75} + \frac{1}{2 \times 25}$

$\boxed{P_T = 375 \text{ torr}}$

(c) P. Range in which both liquid and vapour co exist

37.5 torr to 50 torr

(d)



$x_A P_A^\circ = y_A \times P_T \Rightarrow \frac{n_A}{1} \times 75 = \frac{1-n_A}{1} \times P_T$

$x_B P_B^\circ = y_B \times P_T \Rightarrow \frac{n_B}{1} \times 25 = \frac{(1-n_B)}{1} \times P_T$

$$n_A \times n_B \times 75 \times 25 = (1-n_A) \times (1-n_B) P_T^2$$

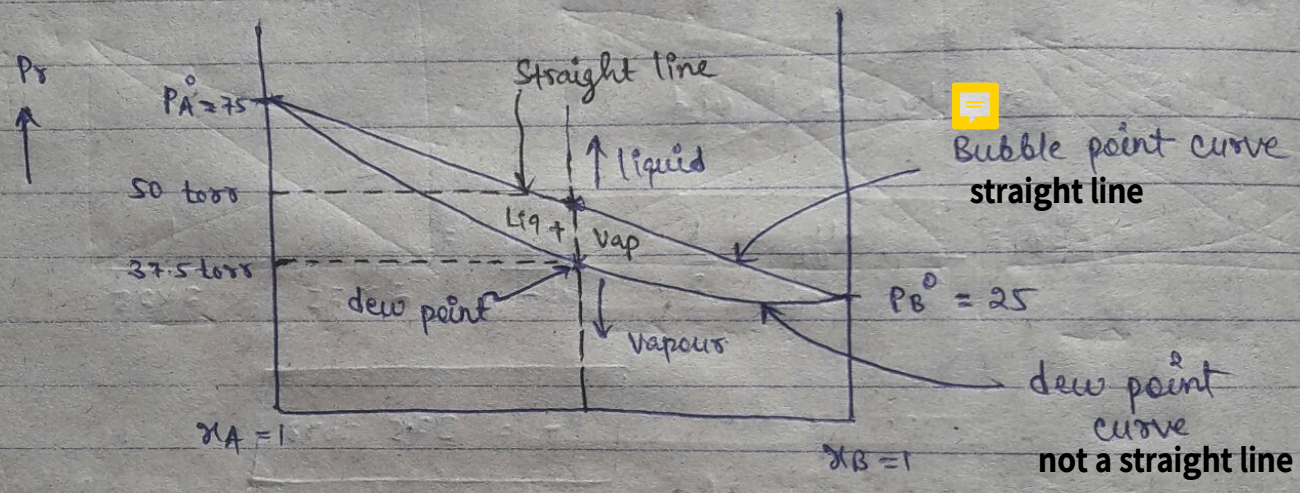
nA = 1 - nB
nB = 1 - nA

$P_T = 25\sqrt{3}$ torr Ans.

'p' = $25\sqrt{3}$ torr

V. Imp

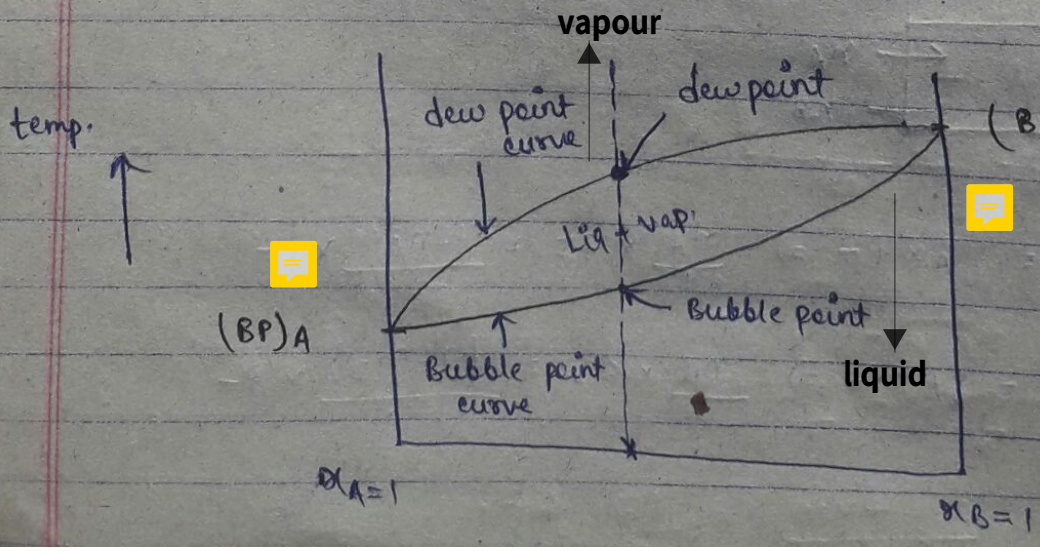
Conclusion : (For ideal Solution)



⟶ Composition.

At constant temperature.

* At a given temperature, a pure liquid has a fixed or definite V.P but an ideal sol^n does not have a fixed V.P



V. imp

→ A pure liquid boils at a fixed temp. at a given pressure but an ideal solution does not boil at a fixed temp.

OR

An ideal solution does not have a fixed boiling temperature

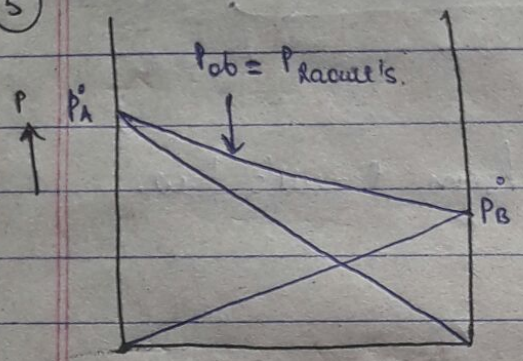
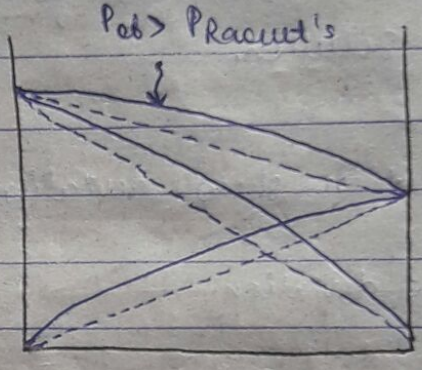
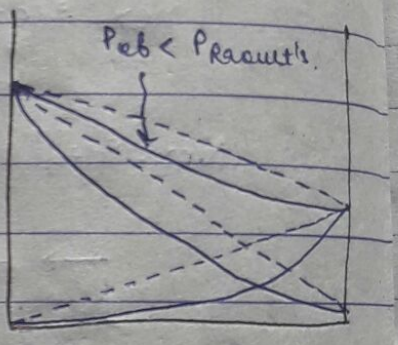
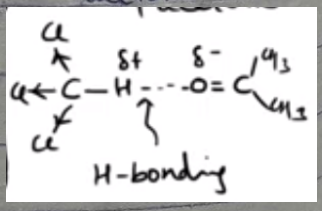
* Non Ideal Solution –
Solution which does not follow Raoult's law is called non-ideal solution.

Non-ideal solutions show deviations from Raoult's law.

→ +ve deviation

→ -ve deviation

Ideal Solution	Non-Ideal solution	
	+ve deviation	-ve deviation
(1) When similar natured liquids mix, A---B forces are similar to that of A---A & B---B forces.	(1) A---B forces are weaker than A---A and B---B forces.	(1) A---B forces are stronger.
(2) $\Delta H_{mix} = 0$	(2) ΔH_{mix} = +ve (endothermic, solution cooling occurs)	(3) ΔH_{mix} = -ve (exothermic solution heats up)

(3) Follows Raoult's law $P_{observed} = P_{Raoult's}$	(3) $P_{ob.} > P_{Raoults}$ $> (x_A P_A^\circ + x_B P_B^\circ)$	(3) $P_{ob.} < P_{Raoult's}$ $\Rightarrow P_{ob} < (x_A P_A^\circ + x_B P_B^\circ)$
	Do not follow Raoult's law.	
(4) $\Delta S_{mix} = +ve$ $\Delta G_{mix} = -ve$	(4) $\Delta S_{mix} = +ve$ $\Delta G_{mix} = -ve$	(4) $\Delta S_{mix} = +ve$ $\Delta G_{mix} = -ve$
(5) 		
(6) Example : → benzene + toluene → n-hexane + n-heptane → Bromobenzene + Iodobenzene	(6) Water + CH_3OH / C_2H_5OH CH_3Cl_3 + CCl_4 Benzene (C_6H_6) + Acetone (CH_3COCH_3)	(6) H_2O + RCOOH (eg CH_3COOH) → H_2O + HNO_3 / H_2SO_4 → Chloroform + acetone
similar natured liquids mixed		 eq. aniline & acetone

Q. 1:1 molar mixture of two volatile liquids A and B (P_A° = 50 torr, P_B° = 100 torr) is taken and observed V.P is 78 torr. Comment what kind of deviation it shows and also comment the change in entropy of solution (system) and surroundings.

Soln $P_{Raoult's} = x_A P_A^\circ + x_B P_B^\circ$

$= \frac{1}{2} \times 50 + \frac{1}{2} \times 100 = 75$ torr

$P_{obs} = 78$ torr $> P_{Raoult's}$

+ve deviation, ΔH_{mix} = +ve (endothermic)

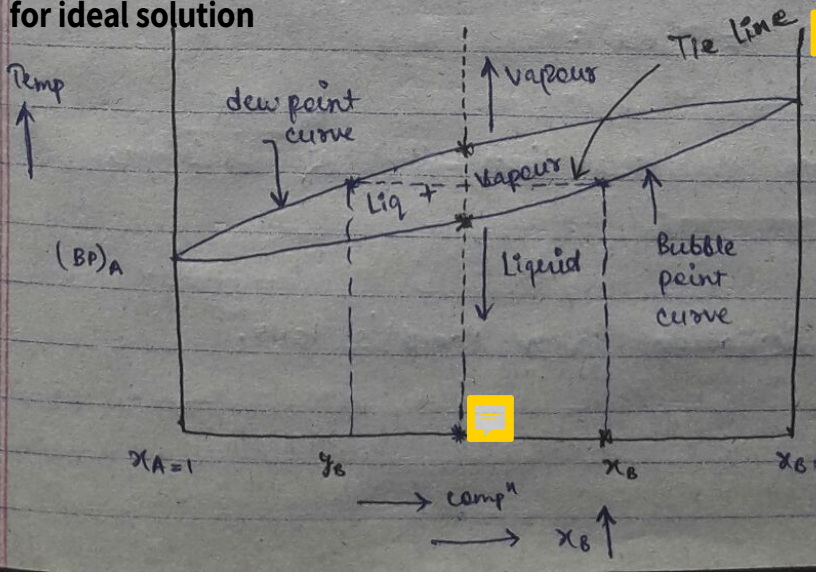
ΔS_{system} = +ve, $\Delta S_{surrounding} = \frac{Q_{surr}}{T_{surr}}$

= -ve

(*) FRACTIONAL DISTILLATION :

A physical method of separation of liquid components of a solution due to difference in their boiling points.

for ideal solution



A → More volatile

B → less volatile

2nd → Essential condition of fractional distillation to occur is that the liquid composition must be different from vapour composition

$$y_A \neq x_A \ , \ y_B \neq x_B$$

→ Distillate or condensate :
When vapour above the liquid are removed and converted into liquid

→ Condensate is richer w.r.t more volatile component.
Residual liquid is richer in less volatile component

2nd * Azeotrope (or Azeotropic mixture) :

→ A non-ideal solution of definite composition which boils at a fixed temp. at a given pressure
i.e Azeotrope behaves like a pure liquid
and vapour composition is same as liquid composition.
Hence, component of azeotropes cannot be separated by fractional distillation.

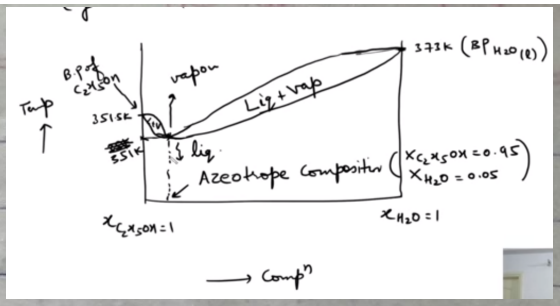
Types of azeotropes :

(a) Minimum boiling azeotrope :

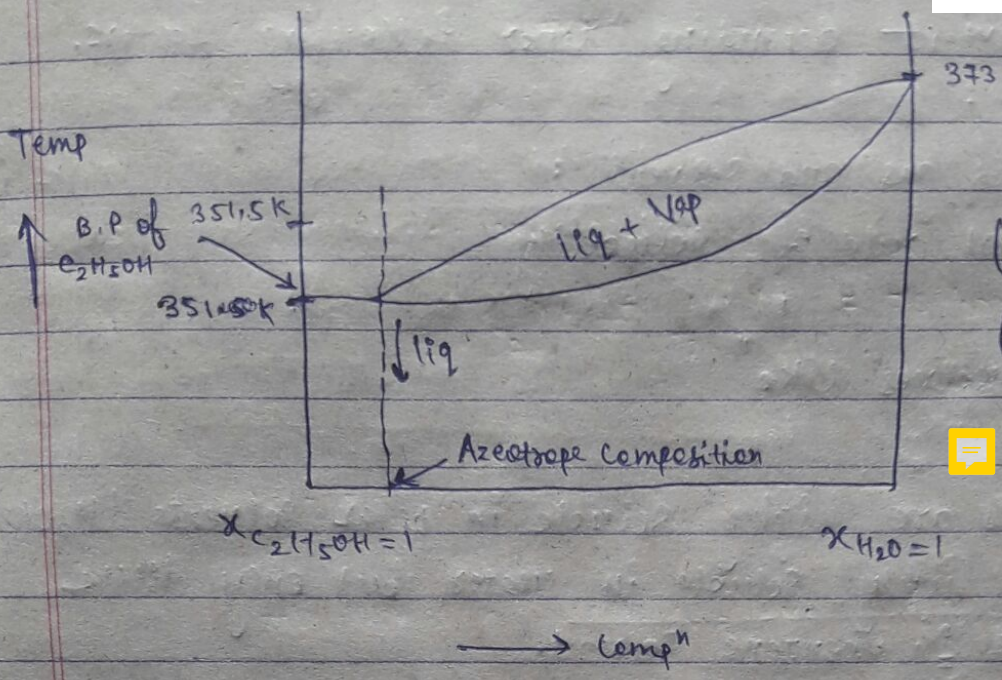
→ B.P of azeotrope is lesser than that of individual components.

→ due to +ve deviation from Raoult's law.

. Any +ve deviation showing solution will form a minimum boiling azeotrope at a particular composition



eg water + ethanol



$(X_{C_2H_5OH} = 0.95)$

$(X_{H_2O} = 0.05)$

need not remember

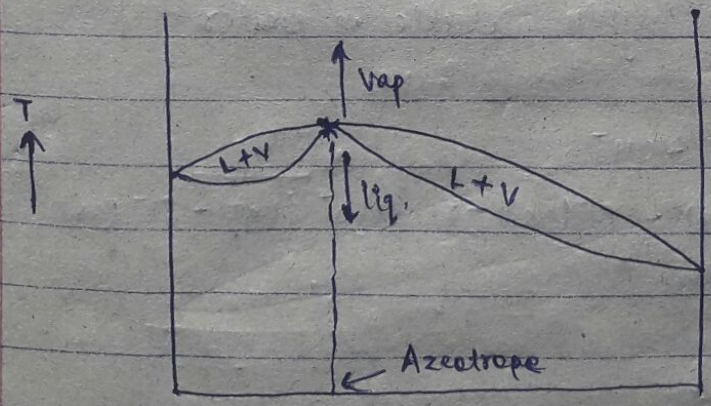
(b) Maximum Boiling Azeotrope :

→ due to -ve deviation

→ B.P of azeotrope is greater than the components.

eg. Chloroform + acetone :

water + CH_3COOH / HNO_3



Solution of gas in liquid :

→ Every gas dissolves in every liquid to a less or greater extent depending upon following factors.

(1) Nature of gas and solvent :
General rule of solubility

"Like dissolves like"

→ Polar gases are highly soluble in polar solvents ($H_2O(l)$)
eg. SO_2 is highly soluble in water but only little soluble in benzene solvent.

eg. Non-polar gases like, CH_4, C_2H_6, BR_2, I_2 vapours etc. are more stable in non-polar solvents like benzene, CS_2, CCl_4, ether etc.

Imp * When a gas undergoes ionisation in a solvent then it is highly soluble in that solvent.

eg. HCl, NH_3 are very highly soluble in water

$$NH_3 \xrightleftharpoons{H_2O(l)} NH_4^+ + OH^-$$

$$HCl \xrightarrow{H_2O(l)} H^+(aq) + Cl^-(aq) \text{ or } H_3O^+(aq) + Cl^-(aq)$$

(2) Temperature :

In general : Dissolution of most solid in liquid ($H_2O(l)$) is endothermic.

→ Dissolution of most gases in water is exothermic.

eg. $X_{(g)} \rightleftharpoons X_{(aq)}$ [exothermic process]

T ↑, Solubility of gas ↓

↑ maximum conc. of gas in given solvent at given temperature.

Claeperon eqⁿ or Clausius - Clapeyron eqⁿ.

$$\ln \frac{C_2}{C_1} = \frac{\Delta H_{solution}}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

where C_1, C_2 are solubility of gas at temp T_1 and T_2.

ΔH_{soln} → Enthalpy of solution.

* Aquatic life (eg fishes) are more comfortable in colder region than the hotter region. as the amount of dissolved oxygen in water decreases in hotter climes

(3) Effect of pressure :

Pressure of Gas ↑ , Solubility ↑

imp → Henry's law : At a given temperature solubility of a gas in a liquid expressed as its mole fraction in liquid is directly proportional to the partial pressure of gas.

According to Henry's law,

$$X_{gas} \propto P_{gas}$$

$$X_{gas} = K \times P_{gas}$$

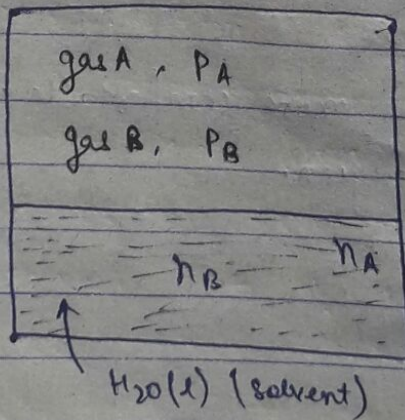
$$P_{gas} = \frac{1}{K} \times X_{gas}$$

$$\boxed{P_{gas} = K_H \times X_{gas}}$$

X_{gas} ← mole fraction

K_H ← Henry's law constant

(unit : Same of pressure unit)



Henry's law

$$P_A = (K_H)_A \times \frac{n_A}{n_T}$$

$$P_A = (K_H)_A \times \frac{n_A}{(n_A + n_B + n_{H_2O})}$$

where n_A = no. of moles of gas A dissolved in water

n_B = no. of moles of gas B ....

→ For low solubility of gases $n_A << n_{H_2O}$

$n_B << n_{H_2O}$

$$P_A = (K_H)_A \times \frac{n_A}{n_{H_2O}}$$

$$P_B = (K_H)_B \times \frac{n_B}{n_{H_2O}}$$

useful for solving Questions.

K_H → Henry's law constant

↳ depends on gas - liquid pair.

ex. different values for solubility oxygen in water & for solubility of oxygen in benzene

$T \uparrow, K_H \uparrow$, Solubility of gas ↓

→ Higher is the K_H value, lower is the solubility of gas at a given pressure.

. The gas which applies greater intermolecular forces with the solvent will be more soluble

→ $He > Ne > Ar > Kr > Xe$

decreasing K_H value or increasing solubility

$H_2 > O_2 > CO_2 > NH_3$

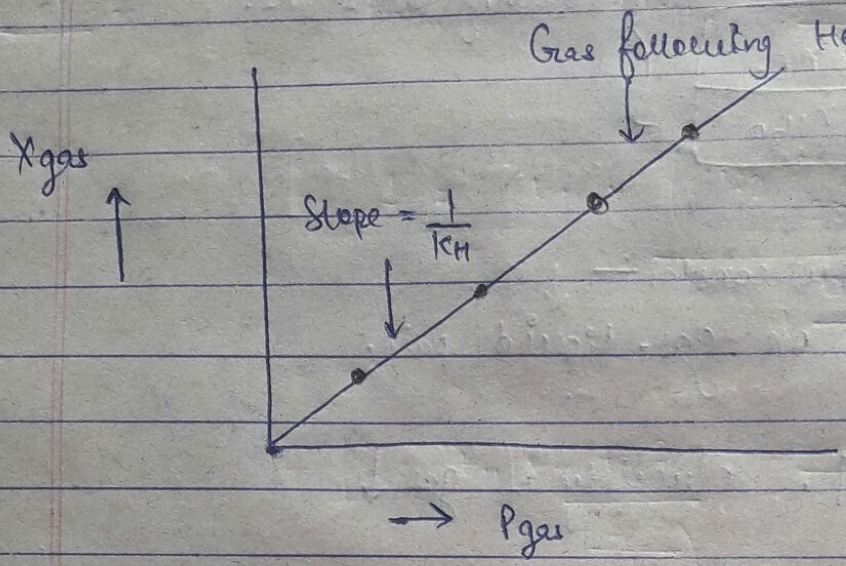
decreasing K_H value.

$CH_4 > C_2H_6 > C_3H_8$

decreasing K_H value

Note: Henry's law is applicable when

(i) Pressure of gas is low.
(ii) Temp. is High.
(iii) Gas should not ionise in water (or solvent)



* Low soluble gases like H_2, He, Ne, O_2, N_2 CH_4 etc follow Henry's law, for water solvent.

* Gases like NH_3, HCl do not follow Henry's law for water solvent.

(*) Applications of Henry's law:

(1) High pressure of CO_2 is maintained while bottling soda/soft-drink.

(2) Decompression sickness or bends:

→ When scuba divers go deep in water, Solubility of N_2 and O_2 in blood and tissues increase. O_2 is well consumed but N_2 accumulates, while coming up towards surface pressure decreases and extra N_2 form bubbles and strikes with tissues or clog blood vessel causing dizziness

pressure increases

exhaustion, pain in muscles and joints called decompression sickness.

→ This problem is minimised by diluting breathing air by He gas i.e N_2 is partially replaced by He gas. He gas is much less soluble in blood and hence lesser bubbles formed and therefore less decompression problem.

Q. What mass of O_2 gas is dissolved in 450 g of liquid water at 300 K temperature if this water is equilibrated with air at 5 atm pressure.

[Air : 20% by vol. of O_2
80% by vol of N_2]

Henry's law constant, $(K_H)_{O_2 - H_2O} = 1.5 \times 10^5$ atm

$(K_H)_{N_2 - H_2O} = 2 \times 10^5$ atm

Soln: $P_{O_2} = (K_H)_{O_2} \times \frac{n_{O_2}}{n_{H_2O}} \Rightarrow$

V. imp

For gas mixture, % by vol = % by mol of a component.

$0.2 \times 5 = 1.5 \times 10^5 \times \frac{n_{O_2}}{\left(\frac{450}{18}\right)25}$

$\Rightarrow \frac{1 \times 25}{1.5 \times 10^5} = n_{O_2} = \frac{W_{O_2}}{32}$

$\Rightarrow W_{O_2} = \frac{32 \times 25}{1.5} \times 10^{-5}$ g Ans

Q. At 300K, 11.2 ml of methane gas (measured at STP) dissolves in 400 ml of water at 1 atm pressure of methane. At what pressure of methane, 10^{-3} mol of CH_4 will dissolve in 500 ml of water at 300 K.

Soln:

$$p_{CH_4} = (K_H)_{CH_4} \times \frac{n_{CH_4}}{n_{water}}$$

$d = \frac{mass}{vol}$

$1\ g/ml = \frac{mass}{400\ ml}$

$mass = 400g$

$$1 = K_H \times \frac{\left(11.2/22.4\times10^{3}\right) mol}{\left(\frac{400}{18}\right)} \quad \text{— (1)}$$

$$p = K_H \times \frac{10^{-3}}{\left(\frac{500}{18}\right)} \quad \text{— (2)}$$

(1) ÷ (2)

$$\frac{1}{p} = \frac{\frac{1}{2\times10^{3}} \times \frac{1}{400}}{\frac{10^{-3}}{500}}$$

$$\frac{1}{p} = \frac{1}{8\times10^{3}} \times \frac{5}{10^{-3}}$$

$$\boxed{P = 8/5 = 1.6 \text{ atm}}$$ Ans.

* Solution of Solid in liquid :-

Colligative Properties :
Property of a liquid solution that depends on total no. of solute particles in solution and does not depend on their nature (i.e size/ shape / charge etc)

* Constitutional property of solution eg. electrical conductance of solution which depends on no. of particles and their nature as well eg. Electrical conductance of NaCl (1M) solution and KCl (1M) are different but colligative property of such solution are same. (Provided both undergo complete Ionisation)

* Four colligative properties are :-

→ lowering in vapour pressure
→ depression in freezing point.
→ elevation in boiling point.
→ Osmotic pressure.

① lowering in vapour pressure :

When a non-volatile solute like salt / sugar etc are dissolved in water (or a volatile solvent) to form a solution, V.P is decreased

P_0 = V.P of pure solvent
P_s = V.P of solution having non-volatile solute

$(P_0 - P_s)$ = lowering in V.P

colligative property.

$\frac{p_o - p_s}{p_o}$ = RLVP (or Relative lowering of V.P)

colligative property

As per Raoult's law,

$p_s = x_{solute} \times p^\circ_{solute} + x_{solvent} \times p^\circ_{solvent}$

p°_{solute} → zero (non-volatile)

$p_s = x_{solvent} \times p^\circ$

$1 - \frac{p_s}{p_o} = (1 - x_{solvent}) = x_{solute}$

$\Rightarrow \boxed{\frac{p_o - p_s}{p_o} = x_{solute} = \frac{n}{n+N}}$

RLVP

$\Rightarrow \frac{n+N}{n} = \frac{p_o}{p_o - p_s}$

$1 + \frac{N}{n} = \frac{p_o}{p_o - p_s}$

$\frac{N}{n} = \frac{p_o}{p_o - p_s} - 1 = \frac{p_s}{p_o - p_s}$

Not RLVP

$\Rightarrow \boxed{\frac{p_o - p_s}{p_s} = \frac{n}{N}}$ → useful for solving question.

$$\frac{p_0 - p_s}{p_s} = \frac{n}{N} \Rightarrow \frac{p_0 - p_s}{p_s} = \frac{n}{\left(\frac{W_{solvent}}{1000}\right)} \times \frac{M_{solvent}}{(1000)}$$

$$\Rightarrow \frac{p_0 - p_s}{p_s} = m \times \frac{M_{solvent}}{1000}$$

m ← molality of solution

$n \rightarrow$ No. of moles of solute

$N \rightarrow$ No. of moles of solvent

NOTE V. Imp

All the colligative properties can be experimentally measured and the chief purpose is to determine molar mass of solute.

Q. In 900ml of liquid water 12g each of urea and glucose are added to form solution. If V.P of water is 25 torr at this temperature, find V.P of solution.

Soln

$$\frac{p_0 - p_s}{p_s} = \frac{n}{N}$$

$$\frac{25 - p_s}{p_s} = \frac{\left(\frac{12}{60} + \frac{12}{180}\right)}{\left(\frac{900}{18}\right)}$$

$$\Rightarrow \frac{25 - p_s}{p_s} = \frac{\left(\frac{1}{5} + \frac{1}{15}\right)}{50}$$

$$\Rightarrow p_s = 25 \times \frac{750}{754} \approx 24.9 \text{ torr}$$

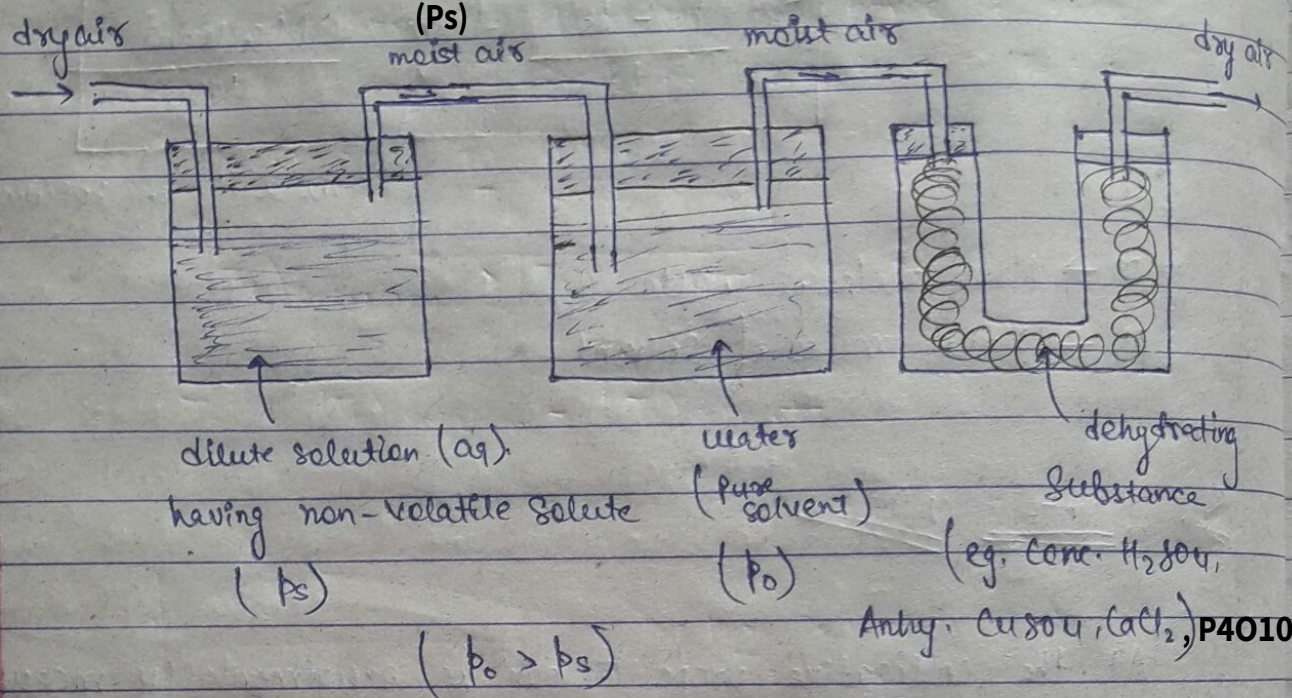
Urea : NH_2CONH_2

Glucose : $C_6H_{12}O_6$

Sucrose : $C_{12}H_{22}O_{11}$

/ measurement

* Experimental determination of RLVP :
(Ostwald & Walker Method)



(P0 > Ps)

V. Imp

⇒ When dry air is passed through a volatile liquid, there occurs a mass loss in liquid which is directly proportional to its V.P

OR

Greater is the volatility of liquid, greater will be the mass loss of liquid when dry air is passed through it.

Mass loss in solution bulb $\propto P_s$

Mass loss in solvent bulb $\propto (P_0 - P_s)$

Total mass loss $\propto (P_s + P_0 - P_s)$

$\propto P_0$

$$\frac{\text{mass loss in solvent bulb}}{\text{Mass gain in dehydrant}} = \frac{(p_0 - p_s)}{p_0} = \frac{n}{n+N}$$

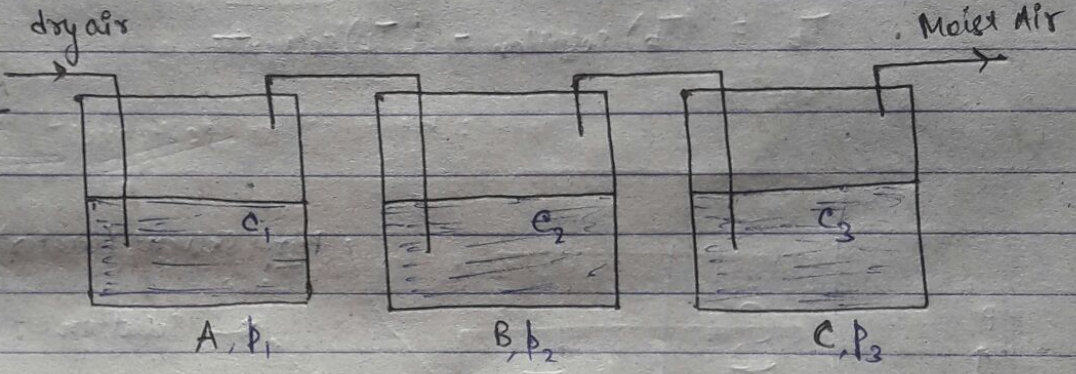
Mass gain in dehydrant = Total mass loss

$\propto p_0$

$$\frac{\text{mass loss in solvent bulb}}{\text{mass gain in dehydrant}} = \frac{W/m_{solute}}{W/m_{solute} + N}$$

From here, we can calculate molar mass of solute, M_{solute}

Question



→ All three solution A, B, C are aq. NaCl solution
→ V.P of solutions are P_1, P_2, P_3
→ conc. of solution are C_1, C_2, C_3
→ Mass loss in A is 2g, mass gain in B is 0.5g
mass loss in C is 1g.

Compare P_1 P_2 P_3 and C_1 C_2 C_3

Soln

When dry air is passed in 'A' → mass loss = 2g
When dry air is passed in 'B' → mass loss = 1.5g
When dry air is passed in 'C' → mass loss = 2.5g

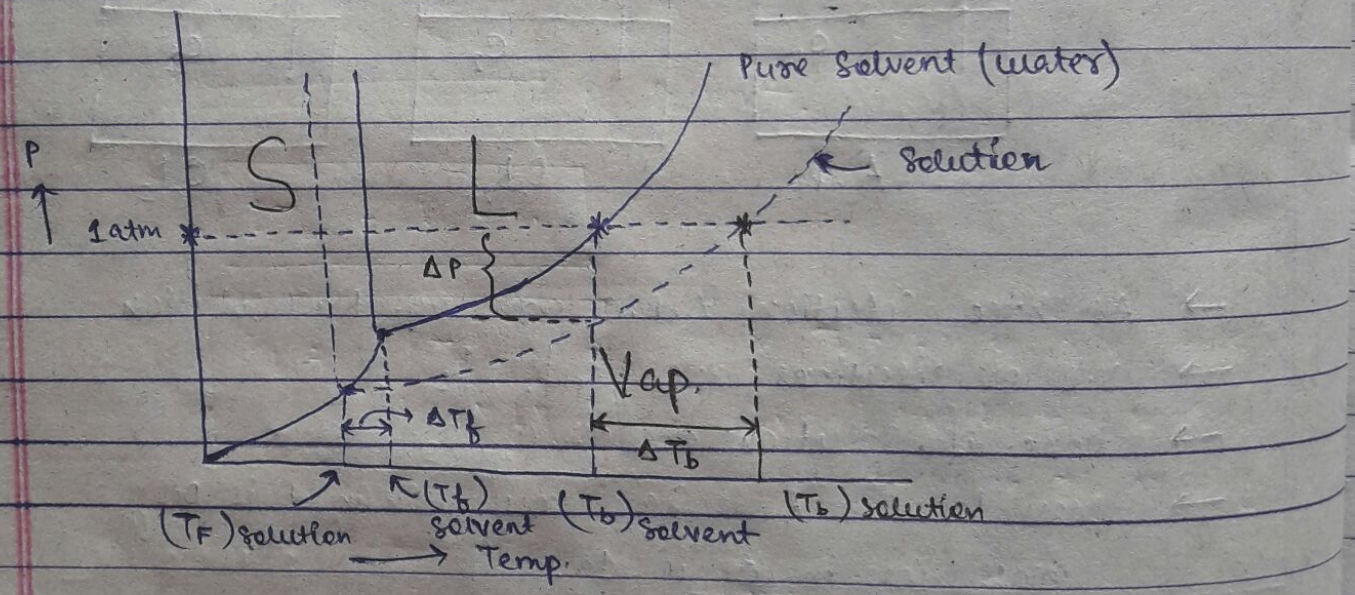
$P_3 > P_1 > P_2$, $C_3 < C_1 < C_2$

(2) Elevation in Boiling Point (ΔT_b)

→ When a non-volatile solute is added into a volatile liquid to form solution, V.P decreases and hence solution need be heated to a higher temp. to boil it so that V.P becomes equal to external pressure.

$$(T_b)_{solution} > (T_b)_{solvent}$$

$$T_b = (T_b)_{solution} - (T_b)_{solvent}$$

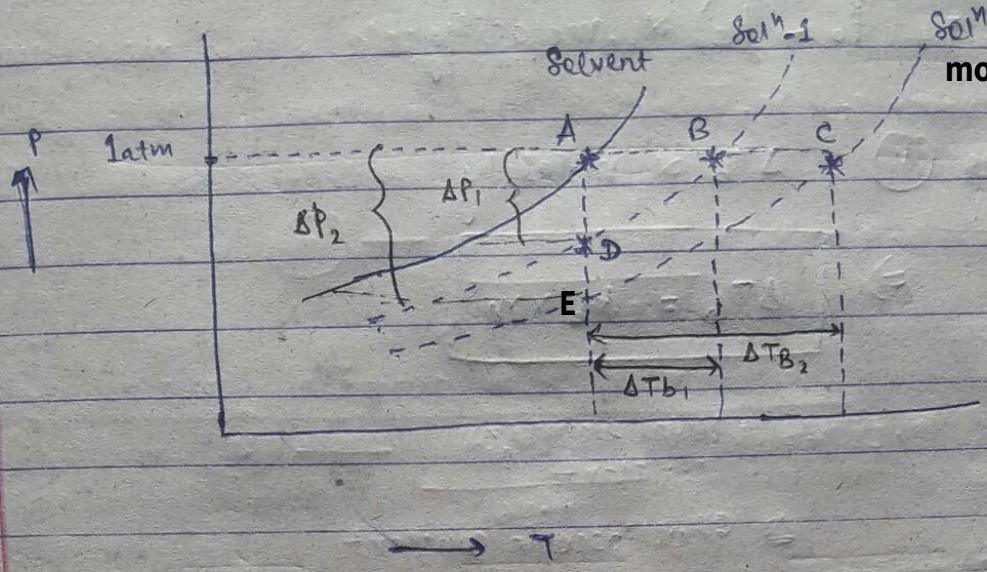


→ On adding non-volatile solute, B.P increases and F.P decreases. In other words, liquid life is more or in greater temp. difference.

ΔT_b – Elevation in BP

ΔT_f – Elevation in F.P

Relation between ΔT_b and molality of solution.



ΔP = lowering in V.P
$= (P_0 - P_s)$

ΔT = Elevation in B.P

→ For dilute solution, segments BD and CE are taken linear.

→ Triangles ABD and ACE are similar.

$$\frac{AD}{AE} = \frac{AB}{AC} \Rightarrow \frac{\Delta P_1}{\Delta P_2} = \frac{\Delta T_{b_1}}{\Delta T_{b_2}}$$

$$\Rightarrow \Delta T_b \propto \Delta P \quad \text{— (i)}$$

$\Delta P = (P_0 - P_s)$ | From Raoult's law

$$\frac{P_0 - P_s}{P_0} = \frac{n}{n+N}$$

For dilute solution

$n << N$

$$\Rightarrow (n+N) = N$$

$$\Rightarrow \frac{P_0 - P_s}{P_0} = \frac{n}{N} = \frac{n}{\left(\frac{W_{solvent}}{100}\right)} \times \frac{M_{solvent}}{1000}$$

$$\Rightarrow (P_0 - P_s) = n \times \frac{M_{solvent}}{1000} \times P_0$$

→ for a given solvent at given temp.

$\Delta P = (P_0 - P_s) \propto$ molality

from eq (i) $\Delta T_b \propto$ molality

$$\Rightarrow \boxed{\Delta T_b = K_b \times m}$$

K_b → Molal elevation constant
or Ebullioscopic constant

Unit : K kg mol^{-1} or °C kg mol^{-1}

* K_b is the property of solvent, constant and given temperature, has nothing to do with the solute

$$\boxed{\Delta T_b = K_b \times m}$$

$$\boxed{K_b = \frac{RT_b^2}{1000 \times L_V}}$$

where T_b = B.P of solvent in K

L_V = latent heat of vaph of liquid in J/g or cal/g

$R = 8.314 \text{ J mol}^{-1} \text{K}^{-1}$
$= 2 \text{ cal mol}^{-1} \text{K}^{-1}$

$$L_V = \frac{\Delta H_{vap}}{\text{Molar mass of solvent}}$$

(3) Depression in freezing point (ΔT_f) :

$\Delta T_f = (T_f)_{\text{solvent}} - (T_f)_{\text{solution}}$

$$\boxed{\Delta T_f = K_f \times m}$$

where K_f = molal depression constant or (cryoscopic constant)
Unit : K kg mol^{-1} or °C kg mol^{-1}

* K_f is the property of solvent.

$$K_f = \frac{RT_f^2}{1000 \times L_f}$$

where K_f = F.P of solvent in K

L_f = latent heat of fusion of solvent in J/g or cal/g

Imp

Question 5g of Sulphur is dissolved in 100g solvent CS_2 and b.p is increased by 0.5°C. Find molecular formula of S in CS_2 solvent.

B.P of CS_2 solvent = 47°C

latent Heat of vaporisation of CS_2 = 80 cal/g

Soln: S_x (sulphur)

$$\Delta T_b = 0.5 = K_b \times \frac{(5/x \times 32)}{0.1\ kg} \quad \left\{ \Delta T_b = K_b \times m \right\}$$

$$0.5 = \frac{RT_b^2}{1000 \times L_v} \times \frac{5}{32 \times 0.1 \times}$$

$$0.5 = \frac{2 \times (320)^2}{1000 \times 80} \times \frac{5}{32 \times 0.1 \times x} \Rightarrow x = 8$$

$\Rightarrow S_8$ Ans

Q. 1g of naphthalene ($C_{10}H_8$) in 50g of a solvent raise the b.p by 0.4°C, 1.2g of an unknown solute in 50g of same solvent raise the b.p by 0.55°C. Find molar mass of unknown solute.

Soln

$$\Delta T_b = 0.4 = K_b \times \frac{(1/128)}{0.05} \quad \text{— (1)}$$

$$= 0.55 = K_b \times \frac{(1.2/M)}{0.06} \quad \text{— (2)}$$

(1) ÷ (2)

$$\boxed{M = 93 \text{ g/mol}}$$

V. Imp

Q. 0.5 mol of a non-volatile solute is dissolved in 1kg of benzene solvent.

(i) Find depression freezing point

(ii) If this solution is slowly cooled down to 275K. find mass of benzene frozen out and also final molality of solution.

Given, k_f (benzene) = 5 K kg mol^{-1}

F.P of benzene = 278.5 K

Soln

(i) $\Delta T_f = k_f \times m$

$= 5 \times (0.5/1\text{kg})$

$= 2.5°C$ or $2.5 K$

$(T_f)_{\text{solution}} = 278.5 - 2.5$

$= 276 K$

Ans

(ii) $\Delta T_f = (278.5 - 275) = 3.5 = 5 \times m'$

$m' = 0.7$ mol/kg

$0.7 = \frac{0.5}{(W_{solvent})_{at\ 275K}}$

$\Rightarrow W_{solvent}$ in liquid form $= \frac{5}{7}$ kg

Mass of benzene frozen $= 1 - \frac{5}{7}$

$= 2/7$ kg Ans.

* Colligative properties when solute undergoes dissociation or association (Vant Hoff factor ('i'))

→ Measured or experimental (or abnormal) C.P is found to be different from expected or theoretical (or normal) C.P if solute undergoes dissociation or association.

$$i = \frac{\text{Experimental (or measured or abnormal) C.P}}{\text{Theoretical (or Normal) C.P considering no dissociation or association}}$$

$$= \frac{\text{Actual no. of moles of solute in solution}}{\text{No. of moles of solute initially taken.}}$$

$$\boxed{i = \frac{M_{normal}}{M_{abnormal}}}$$

(a) Case of dissociation of solute : ($i > 1$)

$$A(aq) \longrightarrow nB(aq), \text{ where } n > 1$$

$t=0$	a	0
$t = eq^m$	$a(1-\alpha)$	$na\alpha$

$$i = \frac{a(1-\alpha) + na\alpha}{a}$$

$$\boxed{i = 1 + (n-1)\alpha}$$

where α = degree of dissociation

n = Sum of st. coeff. of all aq. products per mol of reactant.

eg. $K_4[Fe(CN)_6] \rightarrow \underbrace{4K^+ + [Fe(CN)_6]^{4-}}_{5 \text{ mol}}$

$n = 5$

eg. $2X(aq) \longrightarrow 3Y(aq)$, $n \neq 3$

$n = 3/2$

eg. $2P(aq) \longrightarrow 4Q(aq) + R(s)$

$\boxed{n = 2}$

do not count solids since solids are not part of the homogeneous solution and do not contribute in colligative prop.

(b) When solute undergoes association : (eg dimer, trimer, polymer formation)
$(i < 1)$

$nX(aq) \longrightarrow Y(aq)$, where $n > 1$

$t=0$	a	0	α = degree of association.
$t = eq^m$	$a(1-\alpha)$	$\frac{a\alpha}{n}$	

$$i = \frac{a(1-\alpha) + \frac{a\alpha}{n}}{a}$$

$$\boxed{i = 1 + \left(\frac{1}{n} - 1\right)\alpha}$$

where, n = no. of moles of aqueous reactant converting into 1 mole of product

Note : $i = 1$ for solute that does not undergo dissociation or association.

eg. Urea (NH_2CONH_2), Glucose ($C_6H_{12}O_6$), Naphthalene ($C_{10}H_8$), Sucrose ($C_{12}H_{22}O_{11}$)

Ethylene glycol ($CH_2OH - CH_2OH$)

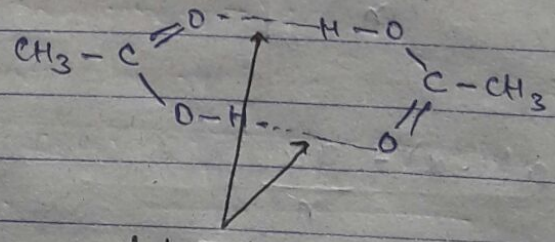
→ Used to clear roads covered with snow. by depression of freezing point, as the freezing point is lowered, hence the snow melts faster

→ Used in radiator of cars as coolant (mixed with water) so that depression in freezing point of water takes place and the water does not freeze easily

NOTE :

CH_3COOH ($HCOOH$, C_6H_5COOH) dissociate in water ($i > 1$) but associate in organic solvents like benzene.

→ CH_3COOH forms dimer through H-bonding in benzene.



Intermolecular H-bonds

Dimer of CH_3COOH or $(CH_3COOH)_2$

Q. Find 'i' in the following

(i) 0.4M aq. Na_2SO_4 ($\alpha = 0.8$)
(ii) 0.5M CH_3COOH in benzene ($\alpha = 0.4$)
(iii) 1M aq. glucose.
(iv) 0.2M NaCl.

Sol^n (1) $n = 3$, $i = 1 + (3-1) \times 0.8 = 2.6$

(2) Dimer formation

$n = 2$, $i = 1 + \left(\frac{1}{2} - 1\right) \times 0.4$

$= 1 - 0.2$

$= 0.8$

(3) $i = 1$

(4) $\alpha = 1$ for NaCl, $i = 2$

⇒ In corporating 'i' in colligative properties,

$$\frac{p_0 - p_s}{p_0} = \frac{in}{in + N}$$

$\frac{p_0 - p_s}{p_0}$ ← experimental RLVP

$$\frac{p_0 - p_s}{p_s} = \frac{in}{N}$$

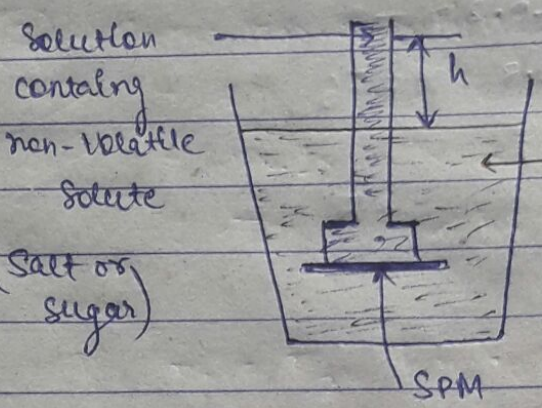
p_s → experimentally observed V.P of solution.

$$\boxed{\Delta T_b = i \times K_b \times m}$$

$$\boxed{\Delta T_f = i \times K_f \times m}$$

* Osmosis :

The spontaneous flow of solvent molecule from pure solvent (or from dilute solution) to solution (or to conc. solution) through semipermeable membrane (SPM).



$\pi = hdg$

↑ density of solution.

SPM : A membrane which allows only solvent molecules to move across it.

eg. Cellophane, cellulose acetate, Parchment
Copper ferrocyanide ($Cu_2[Fe(CN)_6]$)

eg. Animal Bladder (eg. Pig's bladder)

* Osmotic pressure (π) :

An excess hydrostatic pressure developed when pure solvent is separated from solution through SPM at eqm.

OR

The external pressure to be applied on solution so that phenomenon of osmosis is stopped and level of solvent and solution become same, is called osmotic pressure.

* Reverse Osmosis (RO)

When pressure more than osmotic pressure is applied on impure water (or saline water), solvent flows from impure water to pure water, called reverse osmosis.

→ concept of RO is useful in purification of drinking water.

→ Experimentally it is observed that osmotic pressure of a solution at a given temperature is directly proportional to its molar concentration.

$$\pi \propto c$$

$$\Rightarrow \pi = CST$$

where c = molar conc. of solution

S = Molar solution constant

$= 0.0821 \text{ lit atm mol}^{-1} \text{ K}^{-1} \approx \frac{1}{12}$

$= 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

T = Temperature in K.

Using Vant' off factor (i)

$$\boxed{\pi = iCST} = (i_1 C_1 + i_2 C_2 + \dots) \times ST$$

$$\pi = \frac{(i_1 n_1 + i_2 n_2 + \dots)}{V} \times ST$$

↑ Osmotic pressure.

Note : (1) To measure accurate molar mass of bacteria / colloidal particles / polymers / proteins etc. most accurate colligative property is Osmotic pressure measurement for a dilute solution.

(2) Two aq. solutions,

Soln - 1	Soln - 2
π_1	π_2

If $\pi_1 = \pi_2$, then soln - 1 & 2 are isotonic solution.

If $\pi_1 > \pi_2$, solution - 1 is hypertonic
solution - 2 is hypotonic.

(3) Two solutions of similar nature
(solutions having same value of 'i')

Soln - I	Soln - II
π_1, V_1	π_2, V_2

↓ Solutions mixed

$$\pi_f = \frac{\pi_1 V_1 + \pi_2 V_2}{V_1 + V_2}$$

↑ Osmotic pressure of resulting solution.

(4) Osmotic pressure is experimentally measured by Hartley & Berkely method.

not important, need not study

Q. Find osmotic pressure of an aq. solution which is 0.1M each of NaCl and CH_3COOH ($\alpha = 0.1$) at 300K.

Soln: $\pi = (i_1C_1 + i_2C_2) \times ST$

$= (2 \times 0.1 + 1.1 \times 0.1) \times \frac{1}{12} \times 300$ atm

$= 0.31 \times 25$ atm

$= \frac{31}{4} = 7.75$ atm.

For CH_3COOH,
$i = 1 + (2-1) \times 0.1$
$i = 1.1$

Imp

Q. 0.2 lit, 0.1M $BaCl_2(aq)$ is mixed with 0.1 lit, 0.3M $Na_2SO_4(aq)$ solution. Find osmotic pressure of resulting solution in terms of RT.

Soln: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow \underset{\text{white}}{BaSO_4\downarrow} + 2NaCl(aq)$

OR

$Ba^{2+}(aq) + 2Cl^-(aq) + 2Na^+(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4\downarrow + 2Na^+ + 2Cl^-(aq)$

$Cl^- \rightarrow 0.04$ mol
$Na^+ \rightarrow 0.06$ mol

OR

$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow \underset{0.02\text{mol}}{BaSO_4\downarrow}$ (Effective reaction)

0.02 mol (LR) — 0.03 mol − 0.02 = 0.01 mol

$\pi = \frac{i n}{V} \times RT = \frac{0.04 + 0.06 + 0.01}{0.3} \times RT$

$\pi = \frac{11}{30} RT$

Imp

Q 0.1 M urea (aq) , 0.1 M KCl (aq) , 0.1 M Na_2SO_4 (aq) , 0.1 M Na_3PO_4 (aq)
I II III IV

i x c max^n

Select correct option(s)

(A) I > II > III > IV , decreasing osmotic pressure
(B) I < II < III < IV , increasing B.P
(C) I > II > III > IV , decreasing V.P
(D) I < II < III < IV , decreasing F.P

Note : Colligative property should be compared based on 'ic' or 'im' value.

Q. 2.82 g of phenol was mixed with 100g solvent, the freezing point of solvent was depressed by 2.35°c Find % of phenol got dimerised

K_f (Solvent) = 14.1 Kg mol^{-1}

Sol^n $2C_6H_5OH \longrightarrow (C_6H_5OH)_2$

$n = 2$

$$i = \left\{1 + \left(\frac{1}{2} - 1\right) \times \alpha\right\} \Rightarrow i = \left(1 - \frac{\alpha}{2}\right)$$

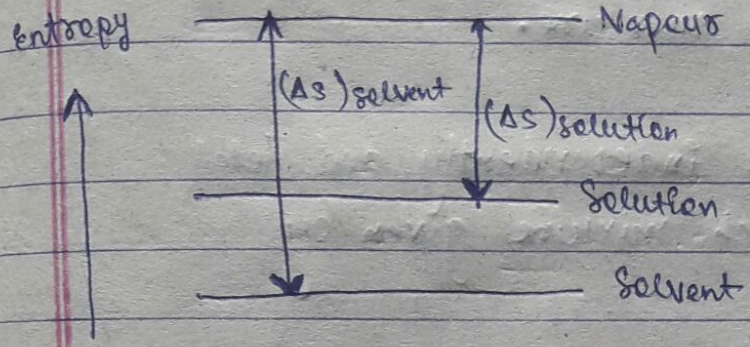
$$\Delta T_f = 2.35 = \left(1 - \alpha/2\right) \times 14.1 \times \frac{\left(\frac{2.82}{94}\right) mol}{0.1 kg}$$

$\alpha = 0.888$

% of phenol dimerised = 88.8 %

(*) Understanding Elevation in B.P or depression in F.P using entropy change :-

→ For Elevation in B.P



For phase change,

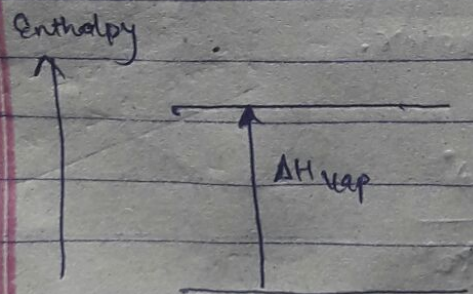
$$\Delta S_{vap} = \frac{\Delta H_{vap}}{T_{B.P}}$$

$$\Rightarrow T_{B.P} = \frac{\Delta H_{vap}}{\Delta S_{vap}}$$

Same for solvent & solution (ideal soln)

$\Delta H_{mix} = 0$

* $(BP)_{solvent} < (BP)_{solution}$.

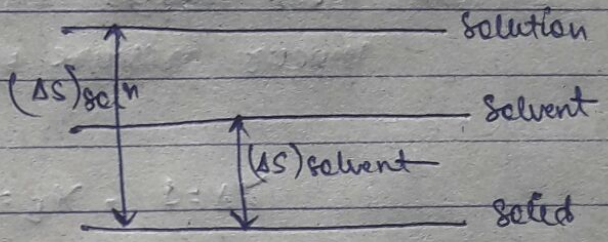


$\Delta H_{mix} = 0$

(For ideal soln)

→ For depression in F.P

$$\Delta S_{fus} = \frac{\Delta H_{fus}}{T_{F.P}}$$



$$T_{F.P} = \frac{\Delta H_{fus}}{\Delta S_{fus}}$$ → same for solvent & solution

* $(FP)_{solution} < (FP)_{Solvent}$.

NOT IMPORTANT

* Immiscible liquid mixture —

→ Some liquid mixture like water + Hg (ℓ) water + benzene / ϕCl / ϕNO_2 etc are considered to be immiscible liquid mixture.

→ Both liquids vapourise independently and total vapour pressure of such mixture given by

$x_A = 1$, $x_B = 1$

$$\boxed{p_T = p_A^\circ + p_B^\circ}$$

↑ V.P of mixture

→ Such mixture boils at a temperature lower than the individual B.P of components. This concept is called steam distillation for separation of liquids like benzene or ϕCl from water.

→ $p_A = p_A^\circ$, $p_B = p_B^\circ$ | $\frac{y_A \times p_T}{y_B \times p_T} = \frac{p_A^\circ}{p_B^\circ}$

$$\frac{y_A}{y_B} = \frac{p_A^\circ}{p_B^\circ}$$

↑ mole fraction of components A & B in vapour at eqm.

$$\left\{\frac{(w_A/M_A)}{(w_B/M_B)}\right\}_{\text{in vapour}} = \frac{p_A^\circ}{p_B^\circ}$$